SCFNet: Semantic Condition Constraint Guided Feature Aware Network for Aircraft Detection in SAR Images

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***Abstract*— Aircraft detection in synthetic aperture radar (SAR) images plays an essential role in satellite observation and military decisions. Due to discrete scattering properties, speckle noise interference, and various aircraft types, many existing methods struggle to achieve the desired detection performance. In this article, we propose an innovative semantic condition constraint guided feature aware network (SCFNet) for detecting different aircraft categories in SAR images. First, considering the discrete scattering properties of aircraft, we design a local–global feature aware module (LGA-M) and a morphological-semantic feature aware module (MSF-M), which can effectively extract the fine-grained feature information contained in SAR images. Second, to effectively fuse different feature information, we construct a feature fusion pyramid (FFP), which uses different branches and paths to reasonably merge multiple feature information types and suppresses background information interference. Third, according to the structure characteristics of aircraft, the global coordinate attention mechanism (G-CAT) is presented to highlight foreground target features and suppress speckle noise interference. Finally, we construct semantic condition constraints, including constraint condition setting, semantic information calculation, and template matching, to improve aircraft localization and recognition accuracy. Extensive experiments demonstrate that the proposed SCFNet can obtain state-of-the-art performance on the SAR aircraft detection dataset, which achieves the average precision (AP) and the F1 score of 94.83% and 95.58%, respectively. The related implementation codes will be made publicly available at https://github.com/darkseid-arch/AirDetection.**

***Index Terms*— Aircraft target detection, attention mechanism, feature extraction and representation, semantic condition constraint, synthetic aperture radar (SAR).**

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I. Introduction

SYNTHETIC aperture radar (SAR) has the capability of all-time and all-weather imaging, which is widely used in military reconnaissance [1], [2], marine surveillance [3], [4], and Earth observation [5], [6], [7]. As an actual application of SAR image interpretation, SAR automatic target recognition (ATR) technology has made significant progress in the past decades. Many algorithms with better recognition performance are proposed [8], [9], [10], [11], [12]. Different from ship and vehicle targets, aircraft targets have the characteristic of high-valued and time-sensitivity. Rapid and accurate detection of aircraft targets can provide valuable information for modern airport management [13] and military decisions [14]. Benefiting from the accelerated development of SAR imaging technology, high-resolution SAR images are more accessible than before, providing an opportunity to achieve refined SAR aircraft detection.

Traditional SAR target detection algorithms can be classified into three categories: statistical analysis modeling [18], geometric contour feature extraction [19], and wavelet transform [20]. The constant false alarm rate (CFAR) [21] is a typical SAR target detection method based on statistical analysis modeling, which performs statistical modeling of background clutter and determines target position and category information by detecting abnormal clutter distribution. Many CFAR-derived SAR target detection algorithms have been proposed and developed rapidly, such as superpixel-based CFAR (S-CFAR) [22], iterative-censoring CFAR (IC-CFAR) [23], intensity-space domain CFAR (ISD-CFAR) [24], outliers-robust CFAR (OR-CFAR) [25], and nonparametric CFAR (N-CFAR) [26]. The geometric contour feature extraction method mainly uses feature extraction algorithms to describe the physical characteristics of different class targets. The widely used feature extraction algorithms include the Hough transform [27], the active contour model [28], and the horizontal gradient histogram [29]. Wavelet transform can effectively process spatial and frequency domain information, which plays an essential role in SAR target detection. Many related modification methods have been proposed, such as the steerable wavelet transform [30], the wavelet scattering network [31], and the parameterization wavelet transform [32]. However, due to poor robustness and limited feature representation capability, these traditional methods cannot achieve

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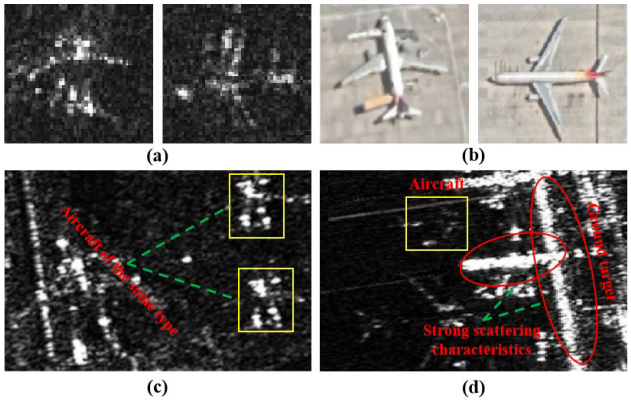

Fig. 1. Aircraft in SAR and optical images. (a) SAR images from Gaofen-3 satellite. (b) Corresponding optical images from Gaofen-1 satellite. (c) Same type of aircraft in SAR images. (d) Interference of background region strong scattering characteristic.

better results for SAR target detection under complex background scattering and discrete scattering distribution scenes.

Benefiting from the powerful feature representation ability of convolution neural networks (CNNs), it has been widely used in many computer vision tasks [33], [34], [35]. The remarkable successes of CNN-based object detectors in optical images provide excellent opportunities for aircraft detection in SAR images. Specifically, Zhao et al. [36] proposed an attention feature refinement and alignment network (AFRAN) composed of the attention fusion module, the deformable lateral connection operation, and the anchor-guided detection module for aircraft detection. To accurately detect aircraft in complex SAR images, Kang et al. [37] proposed a scattering feature correlation network (SFR-Net). SFR-Net uses the scattering point correlation module, the salient fusion module, and the context feature attention to obtain the multiscale feature information. Inspired by the CNN-based object detector, Guo et al. [38] proposed a scattering enhanced attention pyramid network (SEAP-Net), which uses the scattering information enhancement module to enhance the scattering feature representation and uses the multiscale feature pyramid network (FPN) to achieve feature fusion. To obtain the valuable feature information contained in SAR images, Zhang et al. [39] proposed a scattering feature correlation enhancement network (SFRE-Net). SFRE-Net uses a cascade Transformer framework to obtain scattering feature correlation and obtains global features and semantic features by the feature adaptive fusion pyramid and the context attention enhancement module. Based on the discrete scattering characteristics, Zhao et al. [40] proposed the pyramid attention dilated network (PAD-Net), which uses multibranch dilated convolution to obtain the structure feature information and uses the convolution block attention to highlight the foreground target and suppress the background information interference.

Although the above CNN-based detectors have achieved certain results in object detection accuracy, there are still some unsolved problems for aircraft detection in SAR images. Different from optical images, the characteristics of aircraft and background interference in SAR images severely impact the detector's performance. First, as shown in Fig. 1(a), the scattering point distribution of aircraft is more discrete. Due to the existence of strong scattering structures, such as dihedral and trihedral in the aircraft nose (multiple scattering), engine (cavity scattering), and tail (edge diffraction) regions, the scattering point distribution of aircraft is more discrete, and the integrity is weak. However, in the optical images shown in Fig. 1(b), the aircraft geometry and structure information are more obvious and have better integrity. Second, the attitude of aircraft has high sensitivity. Due to the change in the azimuth angle, the angle between the aircraft's strong scattering components and radar wave changes, resulting in the scattering intensity of the same component differences. As shown in Fig. 1(c), two aircrafts of the same type have different scattering intensities. Besides, the strong scattering points of ground things severely impact the scattering characteristics of aircraft targets. As shown in Fig. 1(d), under the influence of the SAR imaging mechanism, the backscattering intensity of the ground target is higher than that of the aircraft to the radar wave; thus, it appears as a highlighted region in the SAR image, which weakens the scattering intensity of aircraft.

To address these problems, we propose a novel semantic condition constraint guided feature aware network (SCFNet) for aircraft detection in SAR images. Different from the direct application of the CNN-based detector to SAR images, SCFNet fully considers the discrete scattering point properties of aircraft, which can extract the fine-grained and nonlinear features, enhance the structure characteristics information, and suppress the background clutter interference. First, to obtain the local and global features, as well as the morphological and semantic features of aircraft targets, inspired by CNN and vision Transformer [41], we design the local–global feature aware module (LGA-M) and the morphological-semantic feature aware module (MSF-M), which can effectively extract and represent valuable feature information. Second, since the existing feature information fusion structure FPN [42] directly merges features of different scales, which causes semantic conflict, SCFNet constructs the feature fusion pyramid (FFP), which can not only realize the reasonable fusion of various scale features but also suppress background clutter interference by using position feature enhancement units (PFEUs). Third, considering the geometric structure characteristics of aircraft, SCFNet uses the global coordinate attention mechanism (G-CAT), which can highlight the foreground target feature and suppress speckle noise interference. Finally, to reduce the false alarm rate (FAR) and improve the positioning and recognition accuracy for different aircraft targets, SCFNet introduces semantic conditional constraints, which enhances detection performance using different constraints. The main contributions of our work can be summarized as follows.

1) Considering the discrete scattering distribution of aircraft and background clutter information in SAR images, this article proposes an end-to-end SCFNet for different classes of aircraft target detection.
2) For scattering points' feature extract, an innovative LGA-M module and an MSF-M module are adopted to extract multiscale feature information. This design solves the problem of SAR image feature extraction.

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3) To achieve multiscale feature information fusion, a novel FFP structure is designed, and the G-CAT is constructed to enhance the saliency of foreground targets and suppress the background clutter interference.
4) Considering the FAR of aircraft detection, we construct the number constraint and spatial position constraint to improve the detector performance for tiny target categories.

II. Related Works

This section introduces related work, including multiscale feature extraction, FPN, and visual attention mechanism in CNN-based object detection.

A. Multiscale Feature Extraction

As an essential component in object detection, the feature extraction and representation effect affects the object detector's performance. In the object detection task, the features to be extracted include local features, detail features, semantic features, and so on. Recently, many feature extraction methods for object detection have been proposed, such as Ma et al. [43] uses multibox, multiscale, and multilevel structures to obtain deep semantic features for object detection. Guo et al. [44] proposed a multiscale feature learning strategy, which uses consistent supervision, residual feature augmentation, and RoI selection for feature extraction and fusion. For global feature extraction, Zhang et al. [45] proposed a multiscale global feature extraction network, which can obtain global and detailed features with different scales. To obtain the multiscale rotation-invariant features, Fu et al. [46] designed a cascade structure for semantic information aggregation, which realized multiscale feature extraction by dense connection and residual learning. Qiu et al. [47] proposed a feature extraction method with adaptive proportional selection, which uses multiple gate units to control the feature map size and uses the scaling factor to complete multiscale feature extraction. To enhance the multiscale feature representation capability, Gao et al. [48] put forward a multiscale residual block, which uses multiple skip connections and feature aggregation modules to extract multiscale features. Although the above methods can achieve multiscale feature extraction, for SAR images, the discrete scattering characteristics of aircraft targets pose a significant challenge to multiscale feature extraction and representation.

B. Feature Pyramid Network

FPN [42] architecture effectively promotes the interaction of cross-scale feature information, which is crucial in the object detection task. Based on FPN, Kim et al. [49] proposed a parallel FPN (PFPNet), which uses spatial pyramid pooling and feature transformation to obtain abstract semantic features and perform feature fusion. To realize multiscale feature information interaction, Zhao et al. [50] designed a graph FPN (G-FPN), where the graph representation learning is used to obtain different node features and uses multitask framework for feature aggregation. For fine-grained feature extraction and representation, Deng et al. [51] proposed an extended FPN, which uses feature transfer and cross-resolution distillation mechanism for feature interaction fusion. To achieve feature extraction and fusion in parallel, Zhu et al. [52] constructed a similarity-based FPN structure, which uses multiscale convolution to extract feature information and uses region merge to fuse different size features. To achieve local and global feature fusion, Qiao et al. [53] proposed a recursive FPN (RFPN), in which the switchable atrous convolution is used to extract local and global features, and uses the switch function to achieve abstract feature fusion. For frequency feature information fusion, Zhang et al. [54] proposed a Laplacian FPN (L-FPN), which uses group cascade convolution to extract different frequency features and uses a multipath structure to achieve feature fusion. Inspired by existing methods, we proposed an FFP for the fusion of local–global features and morphological-semantic features to enhance the feature representation.

C. Visual Attention Mechanism

The visual attention mechanism is essential in obtaining abstract semantic features and performing feature representation. Many attention mechanisms have been proposed for object detection tasks. Li and Wang [55] proposed a hybrid attention mechanism consisting of spatial, channel, and aligned attention to improve single-stage object detector performance. To enhance the feature correlation of different category targets, Tian et al. [56] designed a graph attention mechanism and infers the spatial-semantic relationship by graph siamese architecture. Li et al. [57] constructed a cross-layer attention mechanism, which uses skip connection and feature calibration operations to enhance feature interaction and feature representation for tiny targets. To achieve context feature information aggregation, Miao et al. [58] proposed a cross-context attention mechanism, which enhances the feature representation for target regions by feature reorganization and aggregation. For fine-grained feature extraction, Tian et al. [59] proposed an adaptive attention mechanism combining dilated convolution and self-attention mechanisms to enhance feature extraction performance. To expand the convolution receptive field, Chen et al. [60] constructed a multibranch attention mechanism, which uses parallel spatial and channel attention mechanisms to enhance semantic feature information representation. Based on the above methods, we can see that the attention mechanism is essential for obtaining local feature information and enhancing feature representation.

III. Proposed Method

Fig. 2 shows the overall structure of the proposed detector SCFNet, which is composed of five parts: the backbone network for primary feature extraction, the LGA-M and MSF-M for fine-grained feature extraction, the FFP for multiscale feature fusion, the G-CAT for feature calibration and suppression background clutter interference, and the semantic condition constraint for the final prediction. In this section, we first illustrate the feature aware part for multiscale feature extraction, including local–global feature aware and morphological-semantic feature aware. Next, the pyramid architecture for

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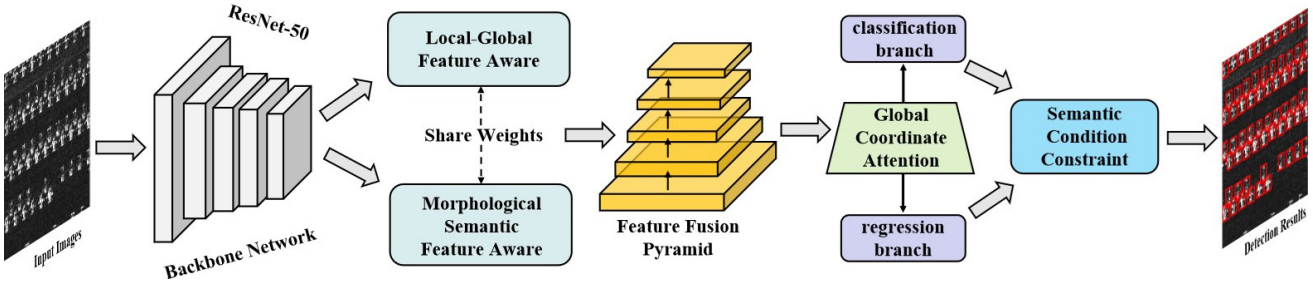

Fig. 2. Overall framework of the proposed SCFNet. This network includes the local–global feature aware, the morphological-semantic feature aware, G-CAT, FFP, and the semantic condition constraint.

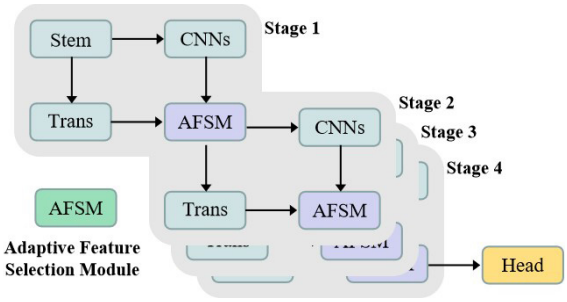

Fig. 3. Structure of LGA-M.

feature fusion is presented in detail based on the inherent attributes of SAR image features. Then, we describe the design of G-CAT for highlighting foreground information and suppressing background noise interference. Finally, we present the constructed semantic condition constraint for generating detection results.

A. Local–Global Feature Aware

The existing CNN model has better local feature extraction capability. Still, obtaining the global feature information in SAR images is challenging due to the limitation of fixed size receptive field range. In contrast, the Transformer model [41] is an effective visual feature extraction framework that can obtain the global feature information of input data.

To obtain the local and global feature information in SAR images, we propose an LGA-M suitable for SAR image feature extraction by combining CNNs and Transformer models. The structure of LGA-M is shown in Fig. 3, which consists of the **Stem** module, the **Stage** feature extraction module, and the **Head** module. Specifically, the SAR image is first preprocessed by the Stem module, then the Stage feature extraction module is used for multiscale feature extraction, and the Head module with deep convolution structure is used to obtain high-level semantic features. In the structure setting of LGA-M, the Stem module uses 1×1 and 3×3 convolutions to perform primary feature extraction on SAR images, and the downsampling operation is used to retain more detail and texture feature information; the Stage module includes the CNN branch and the Transformer branch, in which the CNN branch consists of convolution, batch normalization, and LeakyReLU activation function; and the Head module is composed of four groups CNN modules with different structures, which can effectively fuse the output features of multiple Stage modules.

Since the different computational characteristics of CNNs and Transformer, the direct addition cannot effectively fuse local and global features, which affects the model feature extraction and representation. To achieve the effective fusion of CNNs and Transformer, we propose an adaptive feature selection structure, which uses the attention mechanism to set the spatial gate unit G_S and channel gate unit G_C to complete the selection of spatial and channel information. The structure of adaptive feature selection is shown in Fig. 4, which includes the up-down spatial gate path and the intermediate channel gate path. In the current stage (Stage S), the output of CNNs and Transformer is used as input for the up-down path, and the intermediate features U and V are obtained using the spatial gate unit G_S. The calculation process of the adaptive feature selection structure is given as follows:

$$U/V = G_S(X_C/X_T) \tag{1}$$

where X_C, X_T, U, and V are 3-D arrays. The intermediate path adds X_C and X_T point by point to obtain the fusion feature X and then uses the channel gate unit G_C to obtain decision vectors $\lambda(1-\delta)$ and $\lambda(\delta)$ for selecting channel information. The calculation process is given as follows:

$$\lambda(1-\delta) = (1-\delta)(G_C(X_C + X_T)) \tag{2}$$

$$\lambda(\delta) = \delta(G_C(X_C + X_T)) \tag{3}$$

where δ represents the SoftMax function. The decision vectors $\lambda(1-\delta)$ and $\lambda(\delta)$ are multiplied and added point by point with the intermediate features U and V to obtain the fusion feature Y of the current stage (Stage S). Then, the CNNs and Transformer branches of the next stage (Stage $S+1$) are used for further feature extraction. The calculation process of feature Y is given as follows:

$$Y = U \cdot \lambda(1-\delta) + V \cdot \lambda(\delta) \tag{4}$$

where Y and X_C/X_T have the same size, that is, $Y \in \mathbb{R}^{H \times W \times C}$. The spatial gate unit performs a pooling operation on X_C/X_T in the channel dimension to obtain matrix $H \times W$ and then uses convolution operation to obtain the 2-D weight matrix, and the sigmoid function is used to achieve normalization, and finally, the normalized matrix is multiplied

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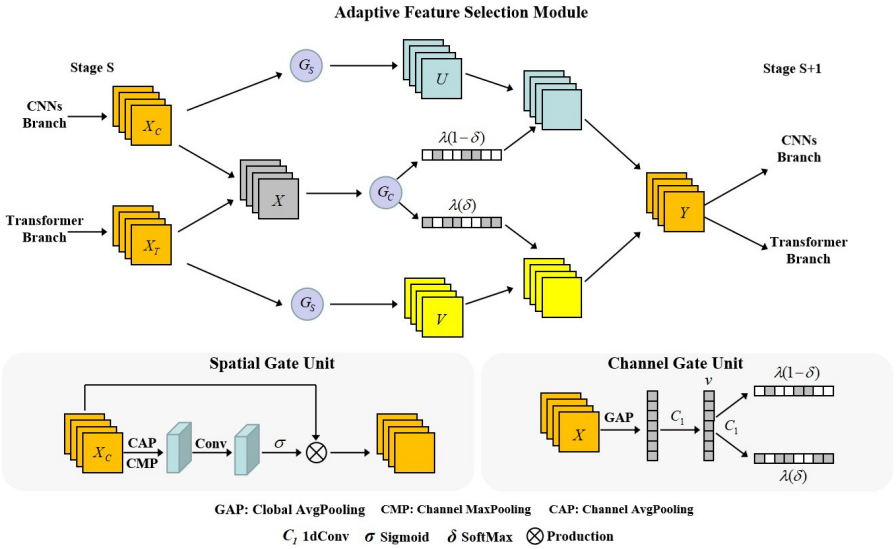

Fig. 4. Structure of the adaptive feature selection module, the spatial gate unit, and the channel gate unit.

by X_C/X_T to obtain intermediate features U and V. The calculation process of the spatial gate unit is given as follows:

$$U/V = X_C/X_T \cdot \sigma\left(C_7\left(P_a^C\left(X_C/X_T\right)\right), P_m^C\left(X_C/X_T\right)\right) \quad (5)$$

where P_a^C and P_m^C represent the average pooling and maximum pooling on channel dimension; C_7 represents the 7×7 convolution operation. In the channel gate unit, it performs the global pooling operation for the feature X on the spatial dimension to obtain a 1-D vector with a size of $1 \times 1 \times C$. To reduce the calculation parameters, 1-D convolution is used to extract the vector feature and obtain the weight vector $\upsilon \in \mathbb{R}^{H \times W \times C}$, which is calculated as follows:

$$\upsilon = C_5^{1d}\left(P_m^S(X)\right) \quad (6)$$

where C_5^{1d} represents 1-D convolution with a kernel size of 5; P_m^S represents the global pooling operation on the spatial dimension. The full connection layer f_c is used to calculate the vector υ to obtain the vectors x and y, and the weight parameters are W^x and W^y. The specific calculation is given as follows:

$$x, y = \upsilon \cdot f_c\left[W^x, W^y\right]. \quad (7)$$

The softmax function is used to normalize x and y so that the sum of elements x^C and y^C at the same position is 1, and then, the decision vectors $\lambda(1-\delta)$ and $\lambda(\delta)$ are obtained by splicing all elements. The specific calculation is given as follows:

$$\begin{cases} y^C &= \delta\left(y^C, x^C\right) \\ x^C &= 1 - y^C \end{cases} \quad (8)$$

$$\begin{cases} \lambda(1-\delta) = \left[x^1, x^2, \ldots, x^C\right] \\ \lambda(\delta) = \left[y^1, y^2, \ldots, y^C\right] \end{cases} \quad (9)$$

where x^C and y^C represent the Cth elements of vectors x and y, which are the same as the number of channels.

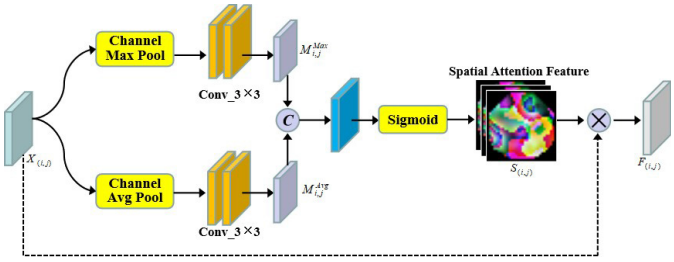

Fig. 5. Structure of the morphological feature aware module.

B. Morphological-Semantic Feature Aware

In SAR aircraft detection, different categories of aircraft have typical morphological features, so the accurate extraction and enhancement of morphological features are beneficial for the aircraft targets' precise positioning and identification. The semantic feature can express the high-level feature information contained in the SAR image and has a powerful discriminate ability for object attributes, so extracting semantic information can further enhance the model feature representation for aircraft objects. Based on the typical characteristics of morphological and semantic features, we propose an MSF-M, consisting of morphological feature aware, semantic feature aware, and morphological-semantic fusion.

1) Morphological Feature Aware: Introducing morphological features into the object detection model is beneficial for improving the object position and recognition accuracy. However, existing methods widely use traditional morphology to preprocess the image; due to the interference of speckle noise, different aircraft objects have high similarity, which makes the conventional morphology unable to extract the complete aircraft contour. To obtain the morphological feature with strong discriminative ability, we constructed a morphological feature aware module suitable for aircraft contour feature extraction, and its structure is shown in Fig. 5. The input of the morphological feature aware module is the output feature

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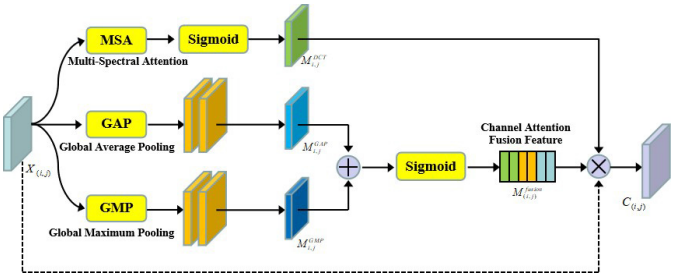

Fig. 6. Structure of the semantic feature aware module.

of local–global feature, which first uses channel maximum pooling and average pooling to downsample and reduce the spatial dimension to obtain feature masks $M_{i,j}^{\text{Max}}$ and $M_{i,j}^{\text{Avg}}$ with the size of $1 \times 2H \times 2W$. The calculation is given as follows:

$$M_{i,j}^{\text{Max}} = \text{Conv_3} \times 3\left(\text{Max}\left(X_{(i,j)}\right)\right) \tag{10}$$

$$M_{i,j}^{\text{Avg}} = \text{Conv_3} \times 3\left(\text{Avg}\left(X_{(i,j)}\right)\right) \tag{11}$$

where Conv_3 × 3 represents the 3 × 3 convolution operation; $X_{(i,j)}$ is the input feature map. Then, the spatial attention $S_{(i,j)}$ is obtained using convolution and concatenation operations, which is calculated as follows:

$$S_{(i,j)} = \delta\left(\text{Conv}\left(\text{concat}\left(M_{i,j}^{\text{Max}}, M_{i,j}^{\text{Avg}}\right)\right)\right) \tag{12}$$

where concat represents the weight connection between feature masks $M_{i,j}^{\text{Max}}$ and $M_{i,j}^{\text{Avg}}$ on channel and spatial dimensions; δ represents the sigmoid activation function. To make the obtained morphological feature have the powerful discriminative ability, the linear transformation operation is used to obtain morphological feature $F_{(i,j)}$ with the size of $C \times 2H \times 2W$

$$F_{(i,j)} = S_{(i,j)} \otimes X_{(i,j)} \tag{13}$$

where $\otimes$ represents the linear transformation operation.

2) Semantic Feature Aware: The semantic feature is typical high-level feature information, and many existing methods widely use dilated convolution to obtain multiscale semantic features and perform feature fusion on channel dimensions. However, these methods ignore the channel correlation, which is essential for SAR aircraft object detection. To fully obtain semantic features and enhance feature correlation of different channel information, we design a semantic feature aware module; the specific structure is shown in Fig. 6. The semantic feature aware uses global average pooling (GAP) and global maximum pooling (GMP) to compress the feature map into a 1-D vector on the spatial dimension; the sigmoid activate function is used to obtain the weight coefficient of different feature maps and adjust the weight parameters in the training process to output the optimal linear combination between channels. As shown in Fig. 6, semantic feature $C_{(i,j)}$ is the fusion of channel attention $L_{(i,j)}$, multispectral attention $M_{i,j}^{\text{DCT}}$, and residual identity mapping $R_{(i,j)}$. Specifically, GAP and GMP first compress the input features in attention masks $M_{i,j}^{\text{GMP}}$ and $M_{i,j}^{\text{GAP}}$ with the size of $C \times 1 \times 1$ on the spatial dimension. The specific calculation is given as follows:

$$M_{i,j}^{\text{GMP}} = \text{GMP}\left(\text{Conv_1} \times 1\left(X_{(i,j)}\right)\right) \tag{14}$$

$$M_{i,j}^{\text{GAP}} = \text{GAP}\left(\text{Conv_1} \times 1\left(X_{(i,j)}\right)\right) \tag{15}$$

where Conv_1 × 1 represents the 1 × 1 convolution operation; $X_{(i,j)}$ represents the input feature map. Then, use channel attention $L_{(i,j)}$ to fuse $M_{i,j}^{\text{GMP}}$ and $M_{i,j}^{\text{GAP}}$

$$M_{(i,j)}^{\text{fusion}} = \sigma\left(M_{i,j}^{\text{GMP}} \oplus M_{i,j}^{\text{GAP}}\right). \tag{16}$$

To enhance the feature representation, a multispectral attention mechanism is introduced to extract feature information from different frequencies, which is calculated as follows:

$$M_{i,j}^{\text{DCT}} = \delta(\text{Freq}) \tag{17}$$

$$\text{Freq} = \text{concat}\left(\left[\text{Fre}q^0, \ldots, \text{Fre}q^j, \ldots, \text{Fre}q^{n-1}\right]\right) \tag{18}$$

where $\text{Fre}q^j$ represents the jth frequency component feature map and feature $M_{i,j}^{\text{DCT}}$ is obtained by weight connection and sigmoid activation function normalization. The multispectral attention $M_{i,j}^{\text{DCT}}$, the channel fusion attention $M_{i,j}^{\text{fusion}}$, and the residual identity mapping feature $R_{(i,j)}$ are fused to obtain the sematic feature $C_{(i,j)}$, which is calculated as follows:

$$C_{(i,j)} = R_{(i,j)} \otimes \left(M_{(i,j)}^{\text{fusion}} \oplus M_{i,j}^{\text{DCT}}\right). \tag{19}$$

3) Morphological-Semantic Feature Fusion: To enhance the feature representation ability, it is necessary to fusion the obtained morphological and semantic features. However, the existing fusion methods use addition and weight connection, which will produce feature information loss and feature dimension mismatch problems. To achieve the effective fusion of morphological and semantic features, we use the backpropagation to dynamically adjust the proportion of morphological and semantic features in the fusion features to construct the fusion feature map adaptively. The specific calculation is given as follows:

$$\text{F_Prop}_{(i,j)} = \left(1 - F_{(i,j)}\right) \otimes X_{(i,j)} \tag{20}$$

$$\text{S_Prop}_{(i,j)} = \left(1 - C_{(i,j)}\right) \otimes X_{(i,j)} \tag{21}$$

$$FM_{(i,j)} = \text{F_Prop}_{(i,j)} \oplus \text{S_Prop}_{(i,j)} \oplus X_{(i,j)} \tag{22}$$

where $\text{F_Prop}_{(i,j)}$ and $\text{S_Prop}_{(i,j)}$ represent the morphological and sematic features adjusted according to the corresponding attention weight coefficients. $FM_{(i,j)}$ represents the final output feature map composed of morphological, semantic, and input features. Since $F_{(i,j)}$ and $C_{(i,j)}$ can be dynamically updated with model training, it can ensure that the fusion feature $FM_{(i,j)}$ has adaptive selectivity.

C. Feature Fusion Pyramid

The existing FPN can achieve multiscale feature information fusion by bottom-up and top-down paths. However, for SAR target detection, due to the interference of complex background and speckle noise, using FPN for local–global and morphological-semantic feature fusion will cause feature information loss. To effectively fuse different feature information of SAR images, we design an FFP, which adds additional local and global features to the pyramid structure for multiscale fusion. The structure of FFP is shown in

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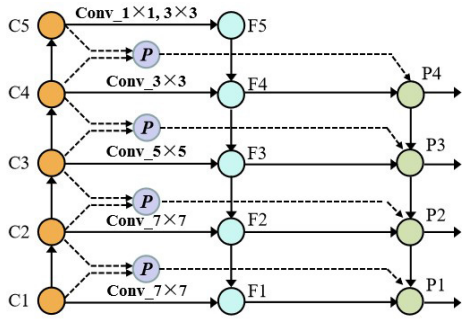

Fig. 7. Structure of FFP.

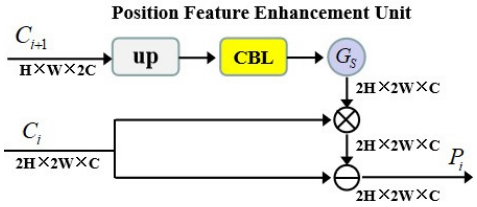

Fig. 8. Structure of PFEU.

Fig. 7, where $C1 \sim C5$ is the output of different feature extraction results in LGA-M and MSF-M, $C1 \sim C2$ is defined as local–global feature, $C3 \sim C4$ is defined as a morphological-semantic feature, and $C5$ is the output of Head module in LGA-M. In FFP, the feature $C5$ is preprocessed by 1×1 convolution to reduce its spatial resolution; then, 3×3 convolution is used to obtain the feature $F5$, and the upsampling operation is used to fuse with feature $F4$. For the local–global feature $C1 \sim C2$, use 7×7 convolution to obtain feature F1 $\sim$ F2. In the feature transmission process, enhancing the object region location information is beneficial to improve the object positioning accuracy. Therefore, we embed PFEU between adjacent features.

The structure of PFEU is shown in Fig. 8, which uses the background features contained in the feature information to highlight the foreground feature representation, so it enhances the object region feature information and outputs it to the prediction unit $P1 \sim P4$. PFEU consists of input units C_{i+1} and C_i, and the output unit P_i, where C_{i+1} represents the local–global feature and $C_{i+1} \in \mathbb{R}^{H\times W\times 2C}$; C_i and P_i represent the morphological-semantic feature; and $P_i \in \mathbb{R}^{2H\times 2W\times 2C}$ and $C_i \in \mathbb{R}^{2H\times 2W\times 2C}$. For input feature C_{i+1}, the spatial resolution is converted to $2H \times 2W$ by using the nearest neighbor interpolation upsampling, and then, the CBL operation is used to convert the channel from $2C$ to C, where the CBL operation is composed of 1×1 convolution, batch normalization, and LeakyReLU. In addition, the spatial gate unit G_S is used to extract the spatial location feature from the output of the CBL module to obtain the feature T_R. The specific calculation process is given as follows:

$$T_R = G_S(\mathrm{CBL}(\mathrm{up}(C_{i+1}))) \tag{23}$$

where up represents the upsampling operation. The sigmoid function is used to normalize the feature T_R to obtain the feature scale function $S(T_R)$ of the foreground object region,

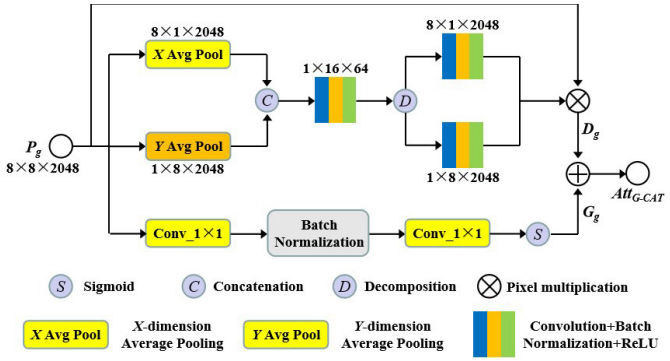

Fig. 9. Structure of G-CAT.

which is calculated as follows:

$$S(T_R) = \frac{1}{1+e^{-T_R}},\ S(T_R) \in (0,1). \tag{24}$$

The scale function is multiplied by C_i point by point to enhance the representation of foreground features, and subtracting it with C_i can eliminate the background region feature. Finally, the output result is multiplied by P_i point by point to obtain the enhanced foreground features. The calculation process is given as follows:

$$P_i^* = (C_i - C_i \cdot S(T_R)) + P_i \tag{25}$$

where P_i represents the original feature and P_i^* represents the enhanced foreground feature.

D. Global Coordinate Attention Mechanism

Because of the speckle noise interference and scattering mechanism influences, the aircraft in SAR images are easily identified as background noise. Although the FFP can alleviate background noise interference, it cannot completely solve the noise interference problem. To make the detection model focus on aircraft feature extraction and suppress noise information interference, inspired by the coordinate attention mechanism (CAT) [61], we propose a G-CAT that can retain the architecture and orientation feature information of the target region while obtaining the channel feature information to improve the aircraft positioning and recognition accuracy.

The structure of G-CAT is shown in Fig. 9, which uses the fusion feature P_g as input and then restructures the weights of feature P_g on the channel and orientation dimensions to output the coordinate attention feature D_g. To reduce the calculation parameters, the feature map size is maintained as 8×8, and the number of channels is 2048. The feature maps' weight reconstruction in different orientations and channel dimensions are composed of coordinate information embedding and coordinate attention generation. The specific operation of coordinate information embedding is to use 8×1 and 1×8 global pooling to decompose the feature map into feature encoder sets on vertical and horizontal dimensions, effectively retaining the space architecture information of aircraft. The calculation is given as follows:

$$z_h = \frac{1}{W}\sum_{0\le j<W} P_g(h,j);\ z_w = \frac{1}{H}\sum_{0\le i<H} P_g(i,w) \tag{26}$$

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where W and H are the width and height of the feature map, and $P_g(i, j)$ is the weight of feature map P_g at the position (i, j); z_h and z_w are coordinate-aware attention maps in vertical and horizontal directions. The coordinate attention generation is to encoder the feature channel and reassigns the target region feature weights. In feature channel encoding, the cascade operation is performed on the encoder feature maps z_h and z_w, and the 1×1 convolution is used to obtain the correlation of different channel information, which is calculated as follows:

$$f = \text{conv_1} \times 1(\text{concat}(z_h, z_w)) \tag{27}$$

where conv_1 $\times$ 1 represents the 1×1 convolution; concat represents the cascade operation; and f is the output attention feature map with spatial and channel features, and its size is $1 \times 16 \times 64$. Then, the attention feature map f is decomposed into the feature map f_h with the size of $8 \times 1 \times 64$ and the feature map f_w with the size of $1 \times 8 \times 64$, and the 1×1 convolution is used to generate the orientation-aware feature map g_h and the position-aware feature map g_w. In feature weight assignment, g_h and g_w act on the feature map P_g by pixel-by-pixel multiplication to obtain the coordinate attention feature D_g. The calculation is given as follows:

$$g_h = \text{conv_1} \times 1(f_h) \tag{28}$$

$$g_w = \text{conv_1} \times 1(f_w) \tag{29}$$

$$D_g = P_g \times g_h \times g_w. \tag{30}$$

For the global information obtaining, we first use 1×1 convolution to reduce the channel dimension of the feature P_g and then use the batch normalization, the 1×1 convolution, and the sigmoid activation function to obtain global information. Moreover, the global feature is added to the coordinate attention feature map to obtain the fusion feature

$$G_g = \sigma\left(\text{Conv_1} \times 1\left(\sigma\left(\text{Conv_1} \times 1\left(P_g(i, j)\right)\right)\right)\right) \tag{31}$$

$$\text{Att}_{G-\text{CAT}} = G_g + D_g \tag{32}$$

where σ is the sigmoid activation function, $\text{Att}_{G-\text{CAT}}$ is the final output feature of G-CAT, D_g represents the coordinate attention feature, and G_g represents the global feature.

E. Semantic Condition Constraint

The purpose of semantic constraints is to use the set condition information as prior knowledge to guide the model training to obtain the optimal target detection results. For the aircraft target detection, we set the constraint conditions as the number constraint of single-class objects and the spatial position constraint of multiclass objects. The semantic condition constraint includes constraint condition setting, semantic information calculation, and template matching.

1) Constraint Condition Setting: The task of aircraft object detection in the SAR image is to detect the aircraft object category and specific location. The semantic constraint of SAR aircraft detection is defined as $U = \{L, F\}$, where L represents the set of object categories and F represents the set of object instances satisfying the constraint conditions. SAR aircraft object detection includes the number constraint of single-class aircraft and the spatial position constraint of multiclass aircraft. The single-class object is the number of same category objects in the SAR image, and its formula is denoted as $|N(L_i)|(\forall L_i \in L)$, that is, the number of category instances L_i can be denoted as $|O(L_i)|(\forall L_i \in L)$. Therefore, the number constraint of the single-class object is defined as follows:

$$\text{F1} = |N(O_{L_i})| \quad \forall L_i \in L. \tag{33}$$

The spatial positions of multiclass objects represent the relative positions of different object instances' center points in spatial coordinates, which are determined by the center point coordinates. The spatial position between different objects can be described by the spatial position relationship between multiple sets of dual objects, so it is only necessary to define the spatial position relationship of a group of object instances. The definition is given as follows:

$$P(O_{L_i}, O_{L_j}) = \langle d^x, d^y \rangle \quad \forall L_i \in L \quad \forall L_j \in L \tag{34}$$

where d^x represents the spatial position relationship between O_{L_i} and O_{L_j} in the horizontal direction. Assume that d^x is the horizontal distance between O_{L_i} and O_{L_j}; $d^x > 0$ means that the center point coordinates O_{L_i} move d^x unit distance on the positive horizontal direction to reach the O_{L_j}; and $d^x < 0$ means that the center point coordinate O_{L_i} moves the d^x unit distance on the reverse horizontal direction to reach O_{L_j}. Similarly, d^y represents the spatial position relationship between O_{L_i} and O_{L_j}; $d^y > 0$ means that the center point coordinate O_{L_i} moves the d^y unit distance on the positive vertical direction to reach O_{L_j}; and $d^y < 0$ means that the center point coordinate O_{L_i} moves the d^y unit distance on the vertical direction to reach O_{L_j}. Therefore, the spatial position constraint of the multiclass object instance is defined as follows:

$$F2 = \left\{P(O_{L_i}, O_{L_j}) | \forall L_i \in L, \forall L_j \in L\right\}. \tag{35}$$

2) Semantic Information Calculation: The position information of aircraft is represented by the center coordinate (x, y) and size (w, h), and the category information is a given specific label. The quantity information of single-class objects can be obtained by counting the object category labels, and the number of the object will be accumulated only the objects that belong to the same category. For example, the number of object instance L_i is calculated as follows:

$$\hat{N}(O_{L_i}) = \sum\left(\text{Detect}\left(\widehat{O}\right) == L_i, L_i \in (0 \text{ or } 1)\right) \tag{36}$$

where $\text{Detect}(\widehat{O})$ represents the category classification result of object instance $\widehat{O}$. The object's relative spatial position is obtained by calculating the center coordinate position of different objects. The absolute value of the coordinate difference in the horizontal direction of the object center point represents the horizontal distance between the different objects. The positive and negative values of the difference represent the relative orientation information on the horizontal direction between objects. Similarly, the absolute value of the vertical coordinate difference between the center points of

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different objects represents the distance between the objects in the vertical direction. The positive and negative values of the difference represent the relative position in the vertical direction between objects

$$\widehat{P}\left(O_{L_i}, O_{L_j}\right) = \left\langle \hat{d}^x, \hat{d}^y \right\rangle = \left\langle \left(x_{O_{L_i}}, -x_{O_{L_j}}\right), \left(y_{O_{L_i}}, -y_{O_{L_j}}\right) \right\rangle. \quad (37)$$

3) Template Matching: The function of template matching is to check whether the object detection result satisfies the constraints in the training stage. Since the constraints that we define are the single-class object number constraints and the multiclass position constraints, it is necessary to construct the corresponding object number matching and object relative position matching rules. The object number matching is to check the consistency between the object numbers of the same category in the object detection results with semantic constraints and the number of categories set by the constraints. The calculation is given as follows:

$$N\left(O_{L_i}\right) == \widehat{N}\left(O_{L_i}\right) \quad \forall L_i \in L. \quad (38)$$

The object spatial position matching is to check the consistency between the positional relationship between the specified objects and the positional relationship set by the constraints, and its purpose is to determine whether the distance between different objects is within the set constraints

$$\text{isEqual}\left(P\left(O_{L_e}, O_{L_j}\right), \widehat{P}\left(O_{L_i}, O_{L_j}\right)\right) = \begin{cases} \text{True}, \left(|\hat{d}^x| \le |d^x| \wedge |\hat{d}^y| \le |d^y|\right) \\ \text{False}, \left(|\hat{d}^x| > |d^x| \wedge |\hat{d}^y| > |d^y|\right). \end{cases} \quad (39)$$

Based on the temple matching rules, only when the detection result is consistent with the set semantic constraints, the model will automatically retrain to obtain the optimal weight until the correct detection result is output. The use of semantic condition constraints can effectively reduce the false alarm in the detection results and improve recognition accuracy.

F. Hybrid Loss Function

The existing object detection loss widely uses CIOU [62] as the loss function, in which the prediction regression error is obtained by calculating the overlap area, the center distance, and the aspect ratio of prediction box P and ground-truth box G. However, the aspect ratio can only reflect the fuzzy difference between the prediction box and the ground truth. To improve the positioning accuracy, we introduce the efficient intersection over union (EIOU) [63] loss function into the SAR aircraft detection model

$$L_{\text{EIOU}} = L_{\text{IOU}} + L_{\text{ctr}} + L_w + L_h \quad (40)$$

$$L_{\text{IOU}} = 1 - \text{IOU} \quad (41)$$

where $L_{\text{ctr}} = (\rho^2(P_{\text{ctr}}, G_{\text{ctr}})/c^2)$, $L_w = (\rho^2(w^{pt}, w^{gt})/c_w^2)$, and $L_h = (\rho^2(h^{pt}, h^{gt})/c_h^2)$; $\rho(\cdot)$ represents the Euclidean distance; P_{ctr}, w^{pt}, and h^{pt} represent the center coordinates, width, and height of the prediction box; G_{ctr}, w^{gt}, and h^{gt} represent the center coordinates, width, and height of the ground-truth box; and c_w and c_h represent the width and height of the minimum bounding detection box. Compared with the CIOU function to calculate the fuzzy error, EIOU uses $(\rho^2(w^{pt}, w^{gt})/c_w^2)$ and $(\rho^2(h^{pt}, h^{gt})/c_h^2)$ to directly calculate the actual error of the width and height in the boundary detection box, which can only improve the positioning accuracy for the object region and can further accelerate the model convergence. The loss function of the proposed SAR aircraft detection model consists of the regression loss L_{EIOU}, the confidence error L^{obj} and L^{noobj}, and the prediction category error L^{cls}, which is defined as follows:

$$L = L_{\text{EIOU}} + L^{\text{obj}} + L^{\text{noobject}} + L^{\text{cls}}. \quad (42)$$

In addition, for SAR aircraft object detection, many object categories only occupy a small proportion of the image, significantly complicating achieving accurate object detection. To improve the detection accuracy for SAR images of small-size aircraft objects, we introduce the focal loss [64] into the constructed loss function, and the calculation is given as follows:

$$L^* = \left(L^{\text{obj}}, L^{\text{noobj}}, L^{\text{cls}}\right) \quad (43)$$

$$L^*_{\text{focal}} = \left(1 + \varepsilon - e^{-L^*}\right)^{\gamma} \cdot L^* \quad (44)$$

$$L_S = L_{\text{EIOU}} + L^{\text{obj}}_{\text{focal}} + L^{\text{noobj}}_{\text{focal}} + L^{\text{cls}}_{\text{focal}} \quad (45)$$

where $\gamma = 2$, ε represents the minimum positive number to prevent the error value of 0, and L_S is the complete form of the constructed loss function.

IV. Experiments and Analysis

Extensive experiments are carried out on the SAR aircraft detection dataset to analyze the effectiveness and robustness of the proposed aircraft detection method. First, we describe the dataset and experimental details. Second, the evaluation metrics are presented. Then, we perform the comparison with the CNN-based object detection methods and illustrate the experimental results. Finally, the ablation experiment and the FAR analysis are implemented, and the function of different compositions is analyzed in detail.

A. Dataset Description

Since there is only one publicly available dataset FG-SAD [65] in the field of aircraft detection in SAR images, we use this dataset for experimental verification and robustness analysis. The SAR images of the FG-SAD dataset are collected from the Gaofen-3 satellite, which contains six different types of aircraft: A220, A330, A320/321, Boeing737-800, Boeing787, and ARJ216. In addition, some uncertain categories are labeled as **other**. However, since FG-SAD contains fewer data samples, to prevent overfitting and improve the model generalization ability, we collected ten SAR images from the TerraSAR-X satellite to supplement the FG-SAD dataset. In addition, for the experimental comparison, the Gaofen-3 image can verify the algorithm detection effect for different types of SAR aircraft targets. The TerraSAR-X satellite image contains many dense scene regions, so it can verify the effect of the proposed method for SAR aircraft object detection in a dense scene. Some sample images of the experimental dataset are shown in Fig. 10. Some detailed information about the dataset is represented as follows.

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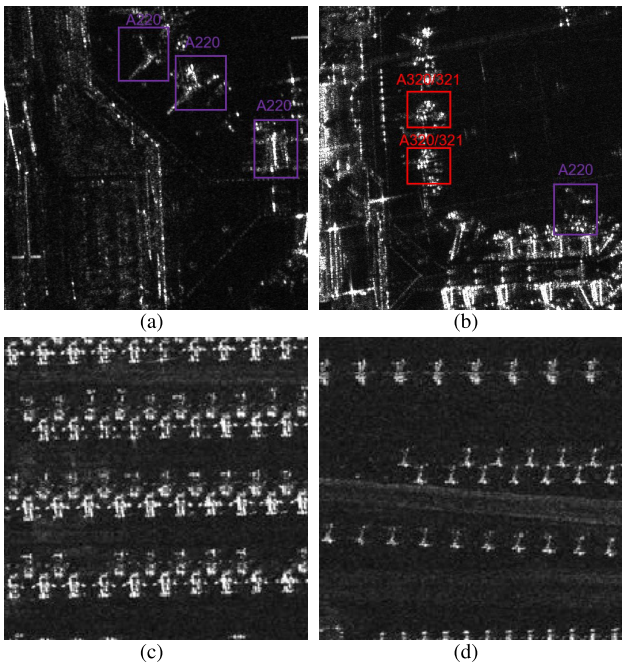

Fig. 10. Illustration of the SAR image dataset. (a) and (b) Subimage from Gaofen-3 satellite used for different types of aircraft detection. (c) and (d) Subimage from TerraSAR-X satellite used for aircraft detection in dense scenes.

1) SAR Images From Gaofen-3: As a C-band spaceborne SAR, the Gaofen-3 satellite can work in 12 different models, including SL, UFC, FSI, and SS. The FG-SAD dataset contains 52 C-band HH polarization images with a 1-m resolution in the SL mode. In the experiment, 3850 nonoverlapped 1024×1024 subimages with 6550 aircraft targets are obtained by cropping the original large images. This dataset covers multiple airports in China, and the images contain much background noise. Fig. 10(a) and (b) shows the subimages from Gaofen-3. In addition, many buildings around the aircraft target bring significant challenges to aircraft detection.

2) SAR Images From TerraSAR-X: TerraSAR-X is an X-band SAR satellite that can work in the spotlight, stripmap, and ScanSAR modes. We collected ten HH polarization images with a 1-m resolution from TerrSAR-X as supplements to the FG-SAD dataset, and the data mainly cover some airports in Germany. The original SAR image is cropped into 512×512 subimgae with 6125 aircraft targets. Fig. 10(c) and (d) shows the subimages from TerraSAR-X. The image from TerraSAR-X has a smaller target size and more complex scenes.

The dataset contains different scenarios, so the size distribution of aircraft targets is relatively uniform, which makes accuracy detection difficult. For further analysis of the dataset, we calculate the area and aspect ratio of the rectangular label box for all aircraft targets and draw the corresponding histogram, as shown in Fig. 11. It can be seen that the aircraft size is diverse, the minimum area covers 527 pixels, and the maximum area includes 34 825 pixels; the rectangle minimum ratio of length/width is 0.31, and the maximum ratio is 3.44 with an average of 0.98.

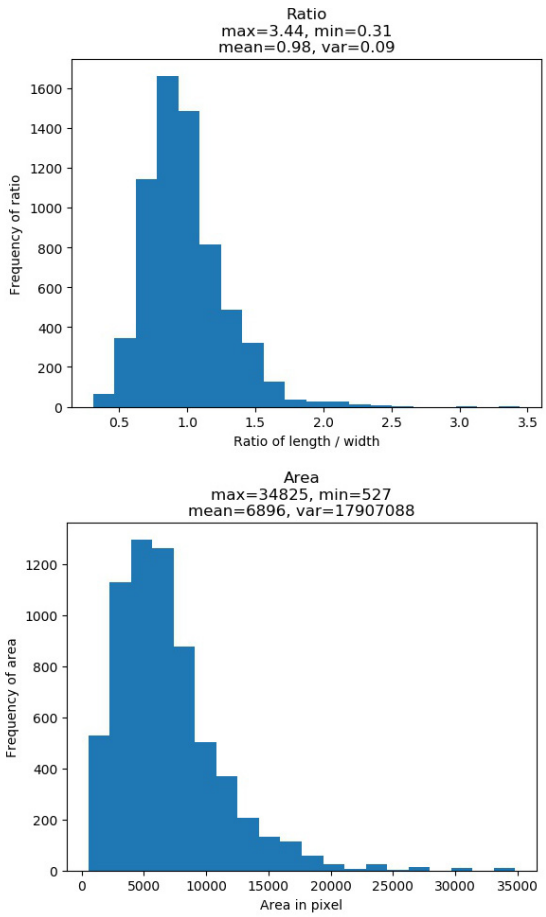

Fig. 11. Statistics of rectangular boxes labeled for aircraft in the dataset.

B. Training Strategies and Implementation Details

The dataset settings and model training hyperparameters during the experiments are described as follows.

1) Dataset Setting: In the experiment, 5100 images are randomly selected from the constructed dataset as the experimental data. The selected images are split into the training set, the testing set, and the validation set according to the ratio of 7:2:1. To improve the generalization performance and prevent overfitting, data expansion methods, including flipping, rotation, scaling, and clipping, are used to augment the training set samples. Moreover, we crop the image to 512×512 pixels in the training process, considering the device memory limitation.

2) Training Strategies: The proposed method is implemented using the Pytorch 1.2.0 framework on a PC running Ubuntu 18.04, CUDA 11.0, and performed training acceleration provided by NVIDIA GeForce RTX 3090 GPU. In the training phasing, ResNet50 [66] is used as the backbone network, and it is pretrained on the ImageNet [67] dataset. All convolution layers are initialized in the same way as [38]. This model is trained using the stochastic gradient descent (SGD) method, and the initial learning rate is set to 0.001.

C. Evaluation Metrics

To quantitatively analyze the performance of the proposed method, we applied some commonly used evaluation metrics,

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TABLE I
PERFORMANCE COMPARISON OF DIFFERENT OBJECT DETECTION METHODS ON THE TESTING SET OF FG-SAD

Models	FAR	Precision	Recall	F1 Score	AP	AP^l	AP^m	AP^s
RetinaNet	0.3283	0.6717	0.6945	0.6829	0.6547	0.4267	0.3873	0.3125
CenterNet	0.2861	0.7139	0.7022	0.7081	0.6832	0.4536	0.3912	0.3248
Cascade RPN	0.2527	0.7473	0.7386	0.7429	0.7156	0.4871	0.4225	0.3354
DR-CNN	0.2138	0.7862	0.7954	0.7907	0.7325	0.4968	0.4387	0.3562
YOLOV5s	0.1987	0.8013	0.8157	0.8084	0.7842	0.5153	0.4625	0.3783
DETR	0.1872	0.8128	0.8245	0.8186	0.7918	0.5627	0.4718	0.3924
TSP-FCOS	0.1685	0.8315	0.8273	0.8293	0.8164	0.5782	0.4863	0.4135
SCFNet (Ours)	0.0469	0.9531	0.9586	0.9558	0.9483	0.5596	0.5374	0.4986

including the FAR, precision, recall, average precision (AP), F1 score, and frames/s (FPS). These evaluation metrics are derived using five established variables, including true positive (TP), false positive (FP), true negative (TN), false negative (FN), and running time for each image. Specifically, the correctly detected aircraft is represented by TP, while FP represents the false alarms. The undetected aircraft is represented by FN, whereas TN represents the correct undetected background targets.

1) False Alarm Rate: The FAR reflects the reliability of the algorithm by calculating the proportion of FP in the overall detection results

$$\mathrm{FAR} = \frac{\mathrm{FP}}{\mathrm{TP} + \mathrm{FP}}. \tag{46}$$

2) Precision: The precision is the proportion of correctly detected targets in the overall detected results

$$\mathrm{Precision} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}}. \tag{47}$$

3) Recall: The proportion of targets that are correctly detected in relation to the overall true targets is defined as the recall

$$\mathrm{Recall} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}}. \tag{48}$$

4) AP: The AP is defined as the area under the precision–recall (PR) curve, and it is used to assess the overall performance

$$\mathrm{AP}_{\mathrm{IoU}} = \int_0^1 P(R)d(R) \tag{49}$$

where IoU is the intersection over the union threshold to determine whether a predicted result is truly positive or FP. In this article, the IoU is set to 0.5. In addition, evaluation metrics AP^s, AP^m, and AP^l from Microsoft COCO [68] are used to refine the detector performance analysis.

5) F1 Score: The F1 score can assess the overall performance of detection algorithms, and it is defined based on the harmonic mean of precision and recall

$$\mathrm{F1\ Score} = \frac{2 \times \mathrm{Precision} \times \mathrm{Recall}}{\mathrm{Precision} + \mathrm{Recall}}. \tag{50}$$

6) FPS: It represents the number of images (frames) processed per second

$$\mathrm{FPS} = 1/T_{\mathrm{sin\,gle}} \tag{51}$$

where $T_{\mathrm{sin\,gle}}$ represents the time taken to process a single SAR image.

D. Comparison to the State of the Art

To demonstrate the advantages of the proposed method, we compared it with seven different state-of-the-art object detection models on the FG-SAD dataset [65]. The comparison methods include CNN-based detectors, RetinaNet [64], CenterNet [69], Cascade RPN [70], Dynamic R-CNN (DR-CNN) [71], and YOLOV5s [72], and Transformer-based detection models, DETR [73] and TSP-FCOS [74]. To ensure the fairness of the experiment, all the compared methods are retrained on the dataset we constructed, and the same training strategy is used as our proposed method. Specific quantitative and qualitative analysis results are given as follows.

1) Quantitative Analysis: The quantitative analysis of different detection models is carried out on the testing set of the FG-SAR dataset. We perform a comparative analysis in terms of detection accuracy and algorithm complexity.

2) Detection Accuracy: The detection accuracy of different methods on the testing set of the FG-SAD dataset is shown in Table I. Compared with other object detection models, our SCFNet has achieved optimal performance in multiple evaluation metrics. Specifically, APs achieved by our method are 16.41% and 13.19% higher than the CNN-based top-place method, YOLOV5s [72], and Transformer-based top-place method, TSP-FCOS [74]. It is obvious that CNN-based methods, RetinaNet [64], CenterNet [69], Cascade RPN [70], and DR-CNN [71], cannot achieve better detection results because the feature information contained in SAR image cannot be fully extracted by simple convolution operation, and the discrete scattering characteristics of aircraft targets reduce the localization and recognition accuracy of the detector. Besides, the F1 scores achieved by YOLOV5s [72] and TSP-FCOS [74] are 14.74% and 12.65% lower than our method, respectively. Benefiting from the powerful global feature extraction ability of the Transformer model, DETR [73] and TSP-FCOS [74] have achieved a slight superiority on AP^l with scores of

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TABLE II
COMPARISON OF CATEGORY DETECTION ACCURACY OF DIFFERENT OBJECT DETECTION METHODS

Categories	RetinaNet	CenterNet	Cascade RPN	DR-CNN	YOLOV5s	DETR	TSP-FCOS	SCFNet (Ours)
A220	0.5381	0.5542	0.5431	0.5784	0.6142	0.6578	0.6857	0.7584
A330	0.5762	0.5678	0.6042	0.6382	0.6783	0.7385	0.7842	0.8341
A320/321	0.6183	0.6387	0.6485	0.6573	0.7152	0.7862	0.8215	0.8672
Boeing737-800	0.6462	0.6658	0.6876	0.6742	0.7324	0.8127	0.8357	0.8863
Boeing787	0.6872	0.7142	0.7312	0.7538	0.7853	0.8536	0.8743	0.9215
ARJ21	0.4875	0.5186	0.5237	0.5447	0.5618	0.6247	0.6385	0.7258
mAP	0.5923	0.6098	0.6231	0.6411	0.6812	0.7456	0.7734	0.8322

TABLE III
COMPARISON OF RUNNING TIME AND ALGORITHM COMPLEXITY OF DIFFERENT OBJECT DETECTION METHODS

Methods	Backbone	Input Size	FPS	Params(M)	FLOPs(G)
RetinaNet	ResNet50+FPN	500×500	73.52	64.28	145.18
CenterNet	Hourglass-52	512×512	142.35	32.75	82.37
Cascade RPN	ResNet50+FPN	832×608	83.28	52.63	123.86
DR-CNN	ResNet101	512×512	11.23	103.81	572.43
YOLOV5s	CSP+Foucus+SPP	608×608	140.61	72.35	165.73
DETR	ResNet101	1024×1024	20.67	63.18	152.31
TSP-FCOS	ResNet101+FPN	608×800	12.64	68.73	135.76
SCFNet (Ours)	ResNet50	512×512	53.62	61.85	151.24

56.27% and 57.82%. However, its ability to identify small and medium aircrafts is still unsatisfactory, e.g., our method is 10.62% and 8.51% higher than these methods on AP^s, and 6.56% and 5.11% higher on AP^m, respectively. It might be because the Transformer model has defects in obtaining refined local feature information, so it cannot correctly identify the boundary information between the aircraft target and the background.

To verify the detection accuracy for different aircraft types, 100 images are randomly selected from the test set as experimental data. Table II shows the detection accuracy of different object detection models for different aircraft categories. As shown in Table II, the mean AP (mAP) of our method has reached 83.22%, which further reflects that the proposed SCFNet has the best experimental performance compared to other object detection methods. For AP scores of specific categories, it can conclude that the detection results from other object detection methods are not ideal for ARJ21 (such as 56.18% of YOLOV5s and 63.85% of TSP-FCOS) and A220 (such as 57.84% of DR-CNN and 65.78% of DETR). In contrast, our method can obtain the AP scores of 75.84% and 72.58% for the A220 and ARJ21. Moreover, compared to all other methods, the proposed method also gains the best experimental result in detecting A330, A320/321, Boeing737-800, and Boeing787. These AP scores related to these objects reached 83.41%, 86.72%, 88.63%, and 92.15%, respectively. The quantitative analysis results of Tables I and II demonstrate the significant advantages of our method in SAR image aircraft target detection in terms of detection accuracy. This is because SCFNet fully considers the SAR imaging mechanism and the discrete scattering characteristics of aircraft targets.

3) Algorithm Complexity: Table III displays the comparison results of detection speed and calculation parameters of different object detection methods. As shown in Table III, benefiting from the lightweight structure design of ResNet50 [66], the proposed SCFNet only takes up 61.85M and 151.24G on parameter volumes (Params) and floating-point operations (FLOPs), which is lower than some CNN-based methods (RetinaNet [64], DR-CNN [71], and YOLOV5s [72]) and Transformer-based methods (DETR [73] and TSP-FCOS [74]). In addition, our method achieves an FPS of 53.62, which is faster than DETR [73] and TSP-FCOS [74] with relatively high detection accuracy, further demonstrating the advantage of the proposed method in terms of real-time performance. However, our method runs slower than some CNN-based methods (RetinaNet [64], CenterNet [69], Cascade RPN [70], and YOLOV5s [72]); this is because multiple submodules in SCFNet for SAR image feature extraction and fusion reduce the model inference speed.

4) Qualitative Analysis: To further illustrate the advantages of the proposed method, PR curves of different compared object detection methods for different object categories are given in Fig. 12. From these PR curves in Fig. 12, it can be seen that these red curves of our SCFNet can provide better detection performance for large-scale aircraft objects (e.g., A330, A320/321, Boeing737-800, and Boeing787) than the other compared object detection models. In addition, our method is more advantageous for tiny object categories

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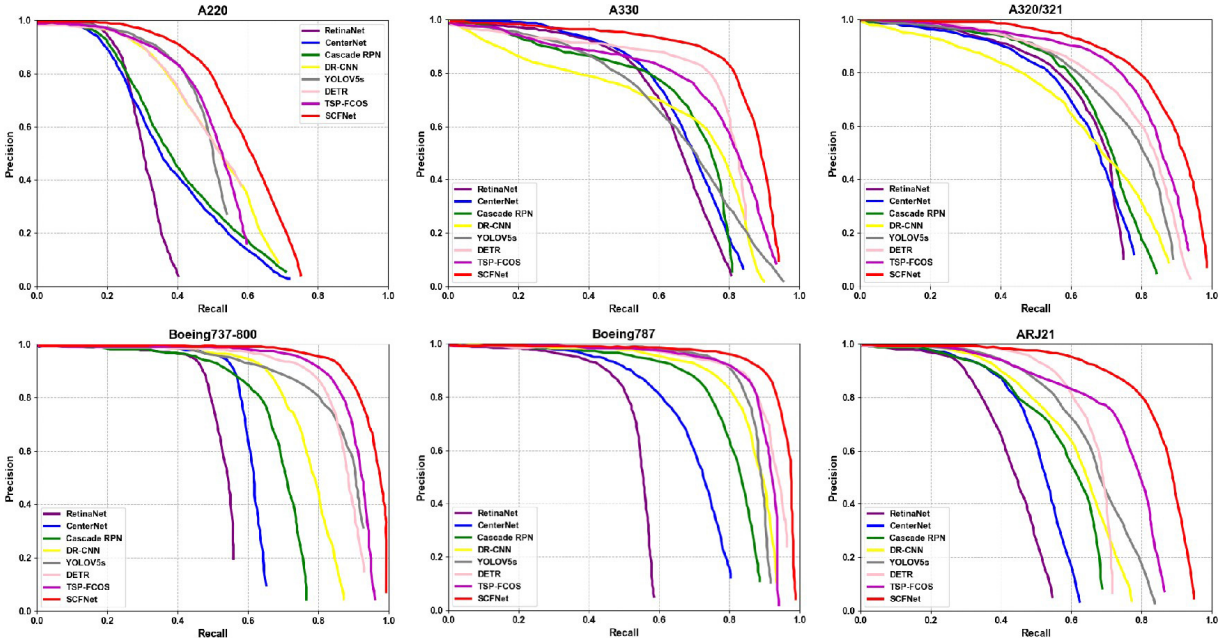

Fig. 12. PR curves analysis on the testing set. The object categories include A220, A330, A320/321, Boeing737-800, Boeing787, and ARJ216.

(e.g., A220 and ARJ21). The comparison results of PR curves show that our method has a better discriminative ability for different categories of aircraft target features in SAR images and can accurately detect and identify aircraft targets.

Fig. 13 shows the visual comparison of different object detection models for aircraft target detection in various scenes. To better demonstrate the robustness and generalization capability of the proposed method, we detect aircraft targets in four scenarios. The validated scenarios include: 1) the aircraft with indistinct backscattering information are densely parked together (in the first and second images); 2) the aircraft target adjacent to ground buildings (in the third and fourth images); 3) strong background noise interference exists due to the complex and angular lounge bridges and other ground facilities (in the fifth and sixth images); 4) low image resolution and interference from complex electromagnetic environments (in the seventh and eighth images). In the first scenario, our SCFNet can accurately detect all aircraft targets and achieve accurate recognition of target category information, while other detection models have missed and false detection problems. For the second scene, the interference information generated by the building has a significant impact on the CNN-based object detection models. For example, the aircraft target adjacent to the building in the third image is missed, while the Transformer-based methods show incorrect target category recognition. For the third scene, the angular lounge bridges and ground facilities produce serious interference information, resulting in missed and false detections for all compared methods, while our method can accurately detect and identify all aircraft targets. In the fourth scenario, the complex electromagnetic environment produces serious speckle noise, which weakens the backscattering information of aircraft targets. For example, in the eighth image, the CNN-based methods cannot accurately obtain the discriminative feature information of the aircraft, so there is a missed detection phenomenon, while the Transformer-based methods have a false detection problem. However, benefiting from the powerful feature extraction capabilities, no false alarms are detected, and FNs are missed by our method.

Except for the above scenario comparison, the detection performance of different object detection models over a dense scene is investigated, as shown in Fig. 14. As shown in Fig. 14, our method can detect all aircraft with fewer false alarms than other compared methods. Specifically, the adjacency phenomenon of aircraft targets makes it difficult for CNN-based methods to extract fine-grained feature information, and the simple feature fusion operation in the CNN model causes the problem of feature information so that more false alarm detection is produced compared with Transformer-based methods. Although the performance of Transformer-based detection models is better than that of CNN-based methods, it is interfered with by the speckle noise contained in the SAR image, and the Transformer model has the disadvantage of insufficient ability to extract global feature information, so some false alarm phenomena occur. For our method, due to the introduction of the fine-grained aware module and the visual attention mechanism, it can detect all aircraft targets, especially for small-scale target categories. The results of aircraft target detection in dense scenes further demonstrate that the proposed SCFNet has strong robustness and generalization capability and can be applied to SAR aircraft target detection in complex scenes.

E. Ablation Studies

In this section, to demonstrate the importance of each component in the proposed SCFNet, we conducted a series of ablation experiments. To make a fair comparison, the same

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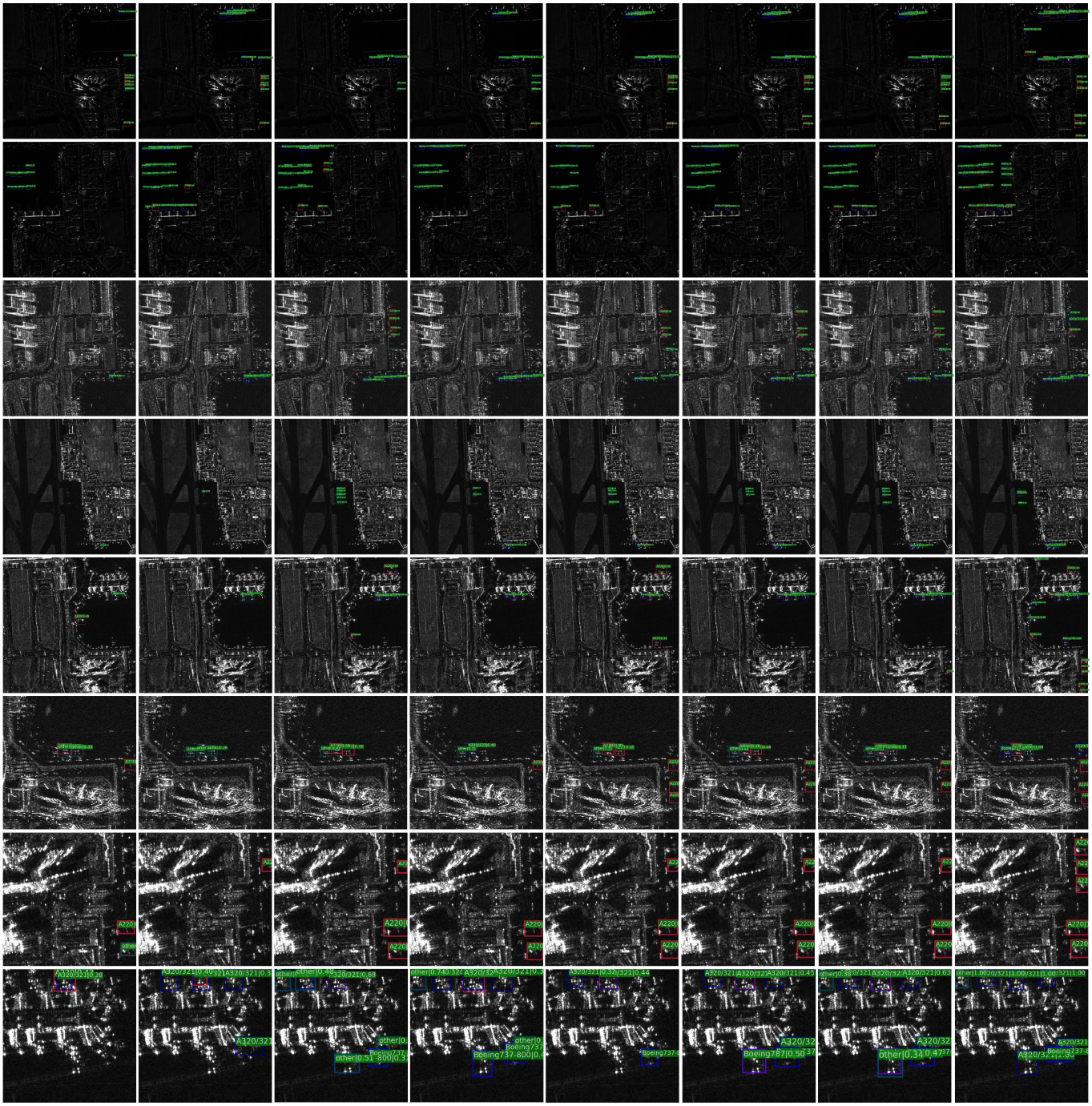

Fig. 13. Visual detection results of different object detection models for aircraft in different scenes.

hyperparameter settings are maintained for all subsequent comparison experiments. The baseline detector is constructed based on the FCOS [35] network, but the feature pyramid structure is removed. The specific comparisons of the experimental results are shown in Table IV, and Fig. 15 displays the comparison of the PR curves of the baseline combining different components and the proposed SCFNet. It can be seen that different designed components have improved the baseline detector performance to a certain extent. Since the proposed SCFNet can fully extract and fuse fine-grained feature information in SAR images, it still has advantages over the structure of the baseline detector combined with different components. As shown in Table IV, compared with the FCOS detector, the baseline combined with all the components designed can improve the AP score by 26.4%, especially for AP^l, AP^m, and AP^s, which can improve 8.44%, 9.19%, and 11.24%, respectively. In addition, these components increase only slightly the computational complexity for model parameters (Params) and FLOPs while improving detection accuracy. The PR curves in Fig. 15 further show that the designed

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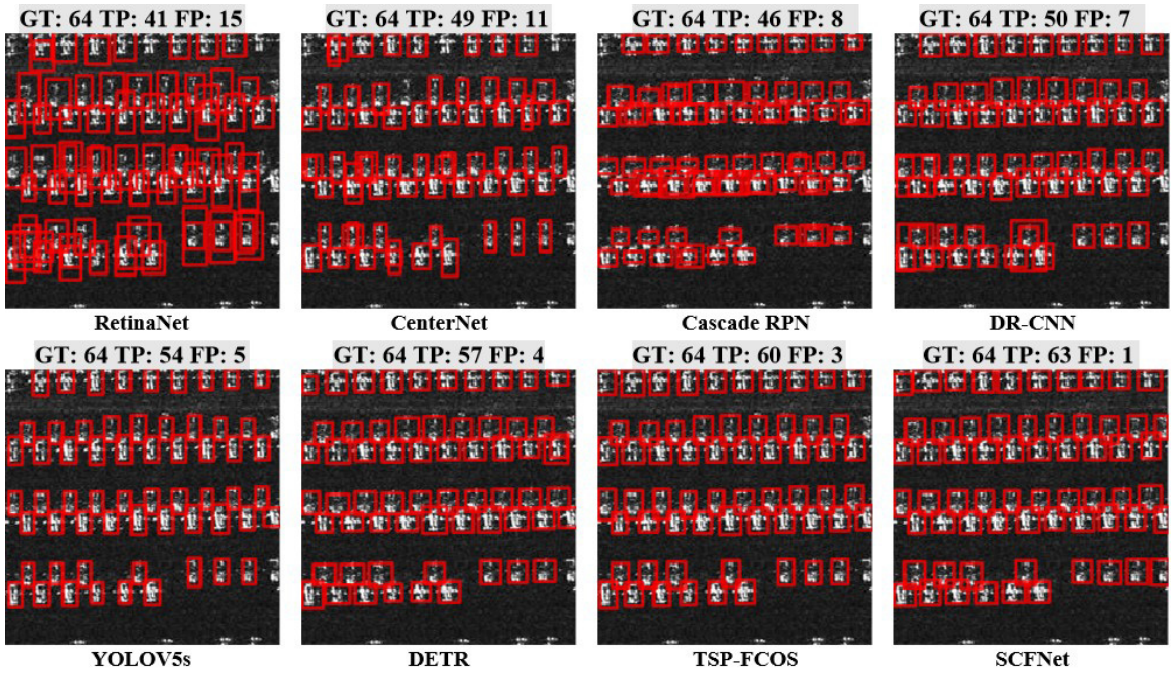

Fig. 14. Aircraft detection results over a dense scene by different object detection models.

TABLE IV
PERFORMANCE OF EACH COMPONENT IN THE PROPOSED SCFNET

Models	LGA-M	MSF-M	FFP	G-CAT	SCC *	AP	AP^l	AP^m	AP^s	FPS	Params(M)	FLOPs(G)
FCOS [35]	-	-	-	-	-	0.6375	0.4032	0.3754	0.3261	72.53	62.73	152.46
Baseline	✓	-	-	-	-	0.7258	0.4172	0.3861	0.3375	70.28	64.87	158.75
	✓	✓	-	-	-	0.7543	0.4286	0.3954	0.3427	68.53	66.52	161.32
	✓	✓	✓	-	-	0.8154	0.4517	0.4238	0.3786	65.42	69.73	164.51
	✓	✓	✓	✓	-	0.8673	0.4625	0.4412	0.3951	62.75	71.25	165.63
	✓	✓	✓	✓	✓	0.9015	0.4876	0.4673	0.4385	59.84	72.14	166.26
SCFNet (Ours)	✓	✓	✓	✓	✓	0.9483	0.5596	0.5374	0.4986	53.62	61.85	151.24

* SCC represents the constructed semantic condition constraint.

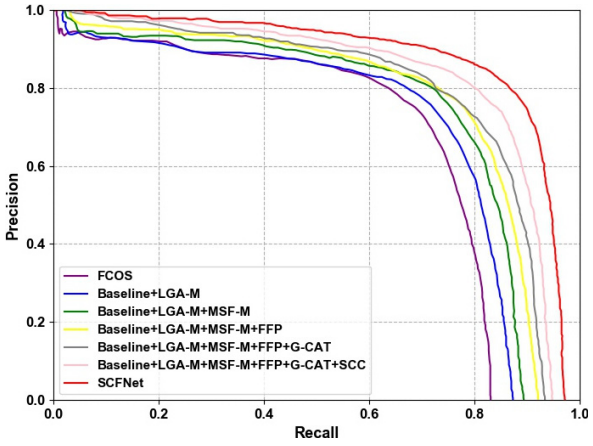

Fig. 15. PR curves of different baselines combined with different components.

different components are conducive to the improvement of model performance. Compared with the FCOS detector, the complete baseline model has obvious advantages.

Next, we will make a detailed experimental analysis of each component designed in SCFNet, including LGA-M, MSF-M, FFP, G-CAT, and semantic condition constraint. The details are given as follows.

1) Effect of LGA-M: Table V demonstrates the effects of three different feature extraction and fusion components in the SCFNet by gradually adding them to the baseline. It can be seen from the ablation experimental results that, with the introduction of LGA-M, all the evaluation metrics gain improvements. To better illustrate the improvement of model performance by LGA-M, we use the CAM [75] to visually compare the feature map of different detectors. The comparison of the detection results is shown in Fig. 16. As shown in Fig. 16(b) and (d), as the baseline model cannot adequately obtain the local and global features of the target area, the classification error and aircraft positioning deviation problems appear in the detection results. In contrast, the proposed LGA-M can enhance the feature extraction and representation ability of the baseline model, as shown in Fig. 16(f) and (h), which improves the classification and positioning accuracy

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TABLE V
ABLATION STUDIES OF LGA-M, MSF-M, AND FFP

Models	LGA-M	MSF-M	FFP	FAR	Precision	Recall	F1 Score	AP	AP^l	AP^m	AP^s
Baseline	-	-	-	0.4785	0.5215	0.5482	0.5345	0.5031	0.3786	0.3518	0.3241
	✓	-	-	0.2517	0.7483	0.7645	0.7563	0.7258	0.4172	0.3861	0.3375
	-	✓	-	0.2418	0.7582	0.7751	0.7665	0.7421	0.4076	0.3957	0.3524
	-	-	✓	0.2349	0.7651	0.7862	0.7755	0.7584	0.4287	0.4186	0.3875

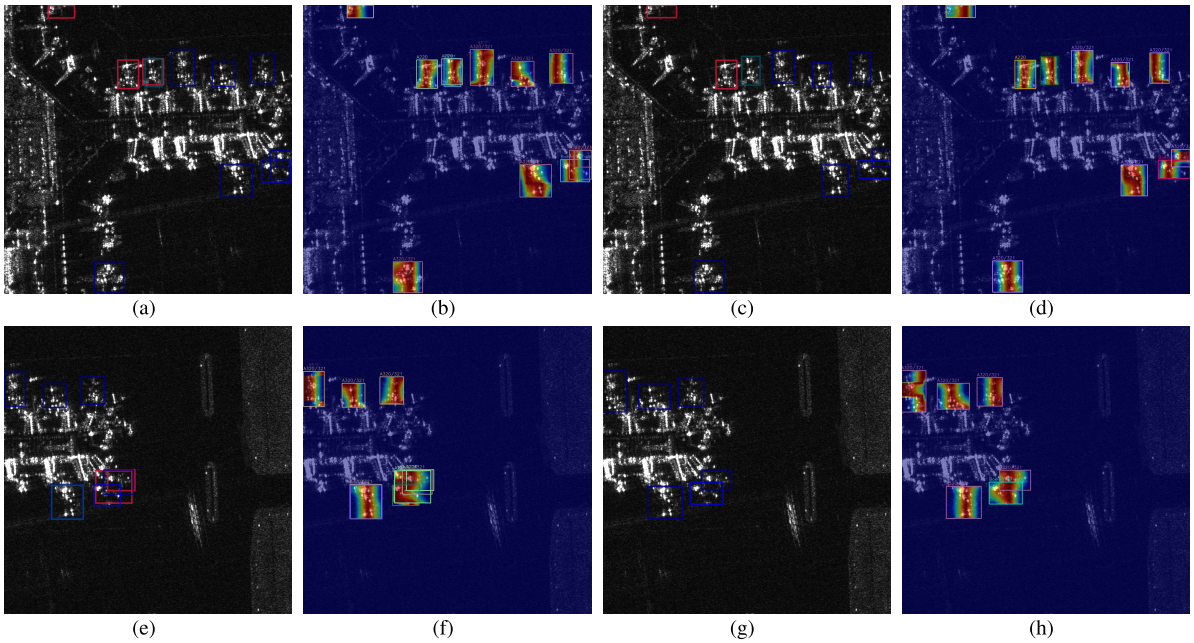

Fig. 16. Detection results of the baseline without and with LGA-M. (a) and (c) Detection results of the baseline. (b) and (d) Visualization of corresponding feature maps. (e) and (g) Detection results of the baseline with LGA-M. (f) and (h) Visualization of corresponding feature maps.

of the baseline model for aircraft targets by fully extracting local and global features. Specifically, the multilayer CNN structure fully extracts the multiscale local features of the target area, while the Transform model obtains the fine-grained global information and improves the feature representation ability of the detector by the fusion of local and global features.

2) Effect of MSF-M: The results in Table V show that the MSF-M increases the F1 score by 23.2% and AP by 23.9% compared with the baseline model. The feature map visualization results in Fig. 17 also demonstrate the superiority of the MSF-M. In Fig. 17, due to the discrete scattering characteristics of the target area, the baseline model cannot accurately obtain the morphological and contour feature information of the aircraft, while MSF-M can focus the model on the target area, weakening the influence of scattering characteristics on the feature extraction process. MSF-M can significantly improve the model's feature extraction and representation ability for aircraft targets in SAR images. The reason is that MSF-M is composed of morphological feature aware and semantic feature aware models. The morphological feature aware enhances the morphological feature in the channel and spatial dimensions, while the semantic feature aware enhances the semantic feature in the global dimension.

3) Effect of FFP: The function of the designed FFP component is to fuse the feature at different scales, which can solve the problem of feature information loss in the feature fusion process of the previous FPN [42] structure. As shown in Table V, the introduction of FFP can increase the AP of the baseline model from 50.31% to 75.84% and the F1 score from 53.45% to 77.55%. The detection results in Fig. 18 further demonstrate the advantages of the FFP structure. As shown in Fig. 18(a), due to the interference of a complex electromagnetic environment, the detection results of the baseline model have the problems of aircraft target positioning deviation, category recognition mistakes, and false alarms. In comparison, the FFP structure improves the detection performance of the baseline model, as shown in Fig. 18(b); it can accurately locate and identify aircraft, and alleviate the interference of speckle noise to a certain extent. FFP can improve the detector performance because its PFEU effectively enhances the ability of the model to extract and represent the position feature information of the aircraft targets.

4) Effect of G-CAT: To further suppress the influence of speckle noise and background clutter on detector performance, we construct the G-CAT. As shown in Table IV, compared with the baseline model with LGA-M, MSF-M, and FFP components, the introduction of G-CAT further improves the

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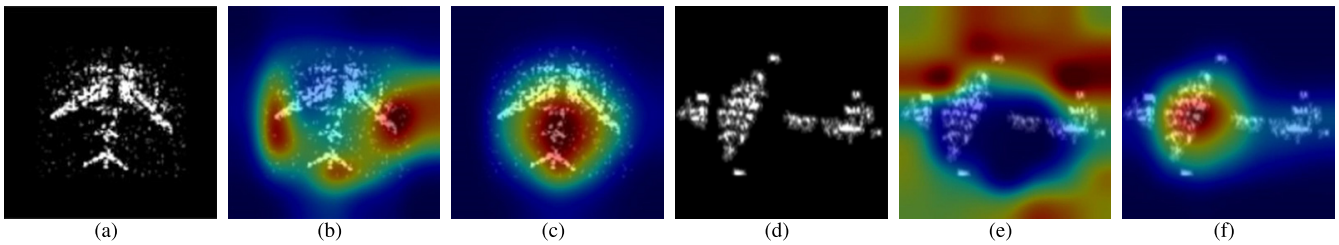

Fig. 17. Comparison visualization of the feature maps. (a) and (d) Original SAR images. (b) and (e) Visualization of the baseline. (c) and (f) Visualization of the baseline with MSF-M.

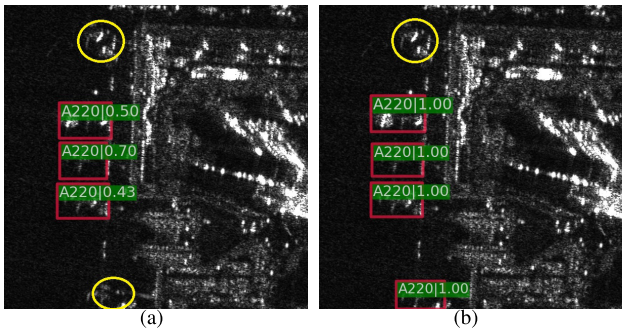

Fig. 18. Detection results of the baseline with and without FFP. The yellow circles represent the missing aircrafts. (a) Detection results of the baseline. (b) Detection results of the baseline with FFP.

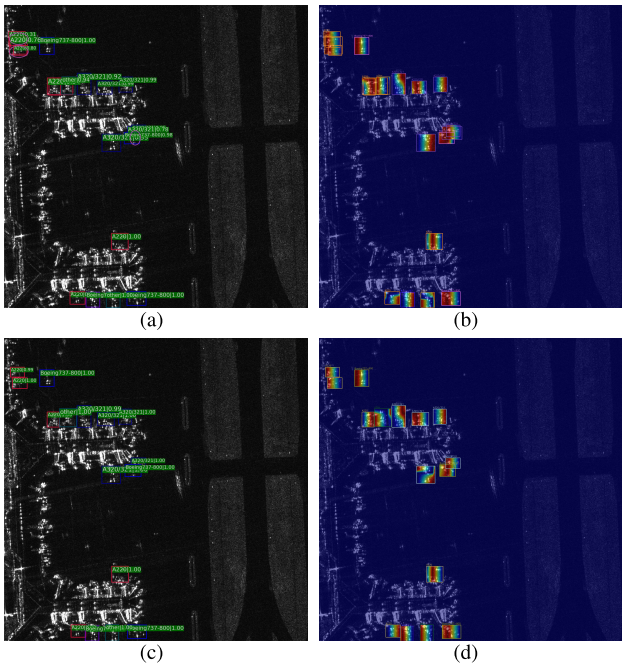

Fig. 19. Detection results of the baseline without and with G-CAT. The purple circles represent the false detection. (a) and (c) Detection results of the baseline and baseline with G-CAT. (b) and (d) Visualization of corresponding feature maps.

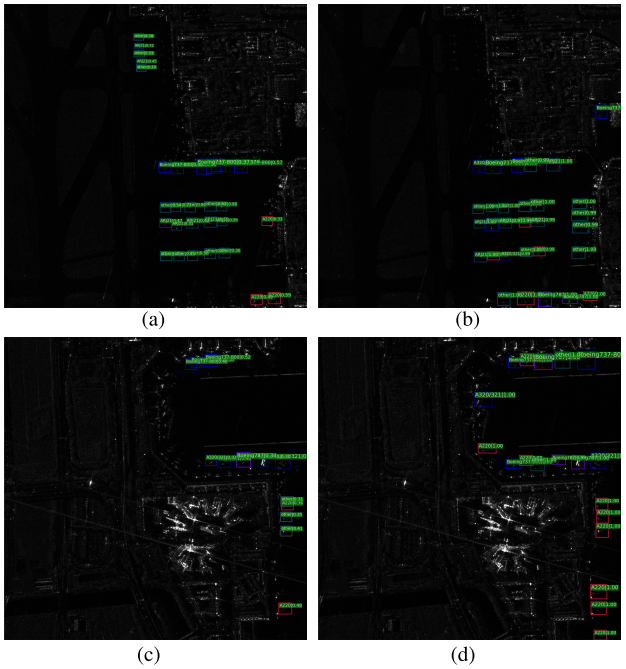

Fig. 20. Detection results of the baseline without and with SCC. (a) and (b) Detection results of the baseline and the baseline with SCC in dense scene targets. (c) and (d) Detection results of the baseline and the baseline with SCC for tiny targets.

detection accuracy with AP^l, AP^m, and AP^s reaching 46.25%, 44.12%, and 39.51%, respectively. The detection results in Fig. 19 also demonstrate that G-CAT can improve the classification and identification accuracy of the baseline model for different categories of aircraft and effectively alleviate the impact of noise on the detector performance. As shown in Fig. 19(a) and (c), the baseline model has category recognition errors and false alarms, while the baseline with G-CAT can correctly identify aircraft categories and with higher confidence scores and positioning accuracy. From the corresponding feature map visualization results, it can be seen that the baseline with G-CAT can focus on the feature extraction of aircraft targets and alleviate the influence of noise on detector performance to improve detection accuracy. The reason is that G-CAT can enhance the feature saliency of the target area from both local and global dimensions, enhance the feature representation of the foreground target, and reduce background interference.

5) Effect of SCC: The purpose of designing semantic condition constraints is to improve the positioning and recognition accuracy of the target area and reduce the false alarm. The SCC that we designed includes the number constraint of the single-class object and the position constraint of the multiclass target. The number constraint can reduce the false alarm of

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object detection in dense scenes, while the position constraint can improve the positioning accuracy for tiny targets. The results in Table IV show that SCC can further improve the detection accuracy of the baseline model, e.g., its AP^l, AP^m, and AP^s are increased by 2.51%, 2.61%, and 4.34%, respectively. The results in Fig. 20 also demonstrate that SCC can improve the detector performance. As shown in Fig. 20(a) and (b), there are some false detections in the results of the baseline model, while the introduction of SCC can reduce the false alarm of the baseline model. Fig. 20(c) and (d) shows that the use of SCC improves the positioning and recognition accuracy of the baseline model for tiny aircraft targets.

In summary, the different components designed in SCFNet can significantly improve the detection performance for different types of aircrafts, suppress the interference of speckle noise and background clutter, and reduce the false alarm to improve the detection accuracy.

V. Conclusion

In this article, a novel SCFNet is proposed for aircraft detection in SAR images by fully considering the SAR imaging mechanism and the discrete scattering characteristics of aircraft targets. Four different components, including the LGA-M, MSF-M, FFP, G-CAT, and the constructed semantic condition constraints, are introduced for adaptive extraction and fusion of local and global features of SAR images, and morphological and semantic features of aircraft targets, reducing false alarms and improving detection accuracy by constraint condition. Ablation studies conducted on the public aircraft detection dataset FG-SAD verified the powerful feature extraction and fusion ability of LGA-M, MSF-M, FFP, G-CAT, and the performance improvements resulting from semantic condition constraint. The test mAP of experiments conducted on the testing set of the benchmark dataset reaches 83.22%, which demonstrates the better detection accuracy and superiority that our method achieves compared with other object detection methods. Furthermore, the correlation between the imaging mechanism of SAR images and the calculation principles of CNN and Transformer should be further considered to design a more reasonable and efficient aircraft detection network.

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